

TOMTABACKEN, Sweden

WMO: 025490

Lat: 57.4981N

Lon: 14.4645E

Elev: 356

StdP: 97.12

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-11.2	-15.6	1.0	-11.4	-13.8	1.2	-9.7	8.8	0.2	8.0	0.9	2.0	10	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	25.2	16.6	23.5	16.1	21.7	15.3	18.3	22.2	17.3	21.1	16.4	20.1	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.6	19.5	15.9	11.8	18.6	14.9	11.0	17.7	53.2	22.5	50.0	21.0	47.1	20.2	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	6.3	5.6	DB	-14.6	27.5	2.8	1.6	-16.6	28.7	-18.3	29.7	-19.9	30.6	-21.9	31.8	(4)
			WB	-15.1	19.8	2.7	1.0	-17.1	20.5	-18.6	21.1	-20.2	21.6	-22.1	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-3.0	-2.6	0.2	5.3	10.1	13.6	15.9	14.9	11.0	6.1	1.9	-1.0	(6)
(7)		DBStd	7.56	4.11	3.95	4.09	3.68	3.77	3.21	3.16	2.65	2.75	3.53	3.60	4.22	(7)
(8)		HDD10.0	1984	402	352	303	146	46	4	0	1	19	127	242	342	(8)
(9)		HDD18.3	4492	661	586	561	389	256	147	91	111	219	379	492	600	(9)
(10)		CDD10.0	559	0	0	0	7	48	112	182	153	51	6	0	0	(10)
(11)		CDD18.3	24	0	0	0	0	1	4	14	5	0	0	0	0	(11)
(12)		CDH23.3	168	0	0	0	0	4	43	96	24	1	0	0	0	(12)
(13)		CDH26.7	16	0	0	0	0	0	2	12	2	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.1	3.1	3.1	3.0	2.9	2.7	2.5	2.6	2.8	3.1	3.4	3.1	(14)
(15)	Precipitation	PrecAvg	808	62	48	43	41	61	81	92	91	75	79	69	65	(15)
(16)		PrecMax	940	122	115	80	100	126	160	198	178	166	171	120	124	(16)
(17)		PrecMin	627	8	7	14	5	20	8	13	18	9	19	25	13	(17)
(18)		PrecStd	93	34	24	18	27	25	37	48	44	43	35	26	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	7.6	13.4	19.3	23.9	26.7	28.1	25.8	21.2	15.6	10.2	8.4	(19)
(20)		0.4%	MCWB	5.8	5.1	7.4	11.4	15.2	15.7	17.4	17.6	15.6	12.4	9.7	8.0	(20)
(21)		2%	DB	4.5	5.1	10.2	16.4	21.7	24.6	25.9	23.3	18.8	13.1	8.8	6.9	(21)
(22)		2%	MCWB	4.1	4.1	5.9	9.5	13.5	15.3	17.2	16.9	14.4	11.8	8.2	6.7	(22)
(23)		5%	DB	3.3	3.8	7.6	13.9	19.4	22.2	24.0	21.4	16.9	11.8	7.8	5.5	(23)
(24)		5%	MCWB	3.0	2.9	4.6	8.0	12.4	14.7	16.6	16.0	13.5	10.7	7.3	5.2	(24)
(25)		10%	DB	2.4	2.6	5.4	11.5	16.9	20.0	22.0	19.6	15.3	10.7	6.8	4.3	(25)
(26)		10%	MCWB	2.0	1.9	3.4	6.8	11.4	13.7	16.1	15.4	12.6	9.8	6.4	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	5.9	8.3	12.2	16.5	18.6	19.6	19.5	16.8	13.2	10.0	8.2	(27)
(28)		0.4%	MCDWB	6.1	6.6	11.7	18.2	21.6	22.7	24.3	23.4	19.8	14.0	10.2	8.3	(28)
(29)		2%	WB	4.2	4.2	6.3	10.3	14.8	16.7	18.6	18.0	15.1	12.0	8.4	6.8	(29)
(30)		2%	MCDWB	4.4	4.7	9.1	14.9	19.5	21.8	23.0	21.1	17.7	12.9	8.7	6.9	(30)
(31)		5%	WB	3.1	3.1	5.0	8.7	13.5	15.3	17.6	16.9	14.0	11.0	7.4	5.3	(31)
(32)		5%	MCDWB	3.3	3.7	7.1	12.5	17.2	20.5	21.8	19.8	16.3	11.7	7.7	5.5	(32)
(33)		10%	WB	2.0	2.0	3.8	7.4	12.1	14.2	16.7	16.0	13.0	10.0	6.5	4.0	(33)
(34)		10%	MCDWB	2.3	2.5	5.1	10.7	15.9	19.1	20.9	18.8	14.8	10.6	6.8	4.3	(34)
(35)	Mean Daily Temperature Range		MDBR	3.7	4.2	6.1	8.3	8.8	8.8	8.1	7.3	6.6	4.8	3.6	3.4	(35)
(36)		5% DB	MCDBR	4.2	5.1	9.3	11.8	11.8	12.2	11.3	9.9	8.7	5.6	4.3	4.3	(36)
(37)			MCWBR	4.3	4.2	5.8	6.6	6.2	5.9	5.0	4.8	5.0	4.2	4.1	4.3	(37)
(38)			5% WB	MCDWB	4.0	4.6	8.5	10.1	10.1	10.5	9.4	8.3	8.0	5.3	4.0	4.4
(39)		MCWBR		4.2	3.9	5.8	6.2	6.1	5.9	5.0	4.8	5.0	4.4	3.9	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.309	0.326	0.362	0.360	0.350	0.372	0.368	0.339	0.329	0.302	0.264	(40)	
(41)		taud	2.291	2.353	2.387	2.358	2.418	2.464	2.415	2.428	2.499	2.455	2.348	2.228	(41)	
(42)		Ebn at Noon	600	742	830	850	877	889	858	837	812	711	561	494	(42)	
(43)		Edh at Noon	47	69	87	104	105	102	105	97	78	63	45	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.37	0.99	2.47	3.95	5.10	5.50	4.97	4.03	2.75	1.27	0.47	0.26	(44)	
(45)		RadStd	0.07	0.19	0.28	0.54	0.49	0.29	0.64	0.41	0.38	0.23	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (61 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	N/A

Nomenclature: See separate page